

Individual Reflection | Software Engineering Project Management

EXECUTIVE SUMMARY

Overall, the module was informative and interesting, leading to good opportunities to place the learning to date in a real world context. In addition, there was good opportunity to revisit the coding skills which sit at the heart of the course, and were the main reason for enrolment to the Masters in Computer Science. In particular, I enjoyed using the linters to modify my own code which deepened my understanding of this area.

There were, however, many barriers to overcome when completing the module, stemming from a lack of prior knowledge or professional experience in the IT sector. However, I tried to overcome these by drawing parallels to my own professional experience of leading projects in my own professional domain, secondary education. While the module may have been well sequenced for those who work in, or have experience of, the IT sector, the technical challenges required additional time and persistence to complete.

Time management was also a specific barrier due to personal reasons. My family are recovering after a house fire and it has been challenging to complete work, study and organise house renovations simultaneously. Nonetheless, this has built my resilience and familiarity with meeting tight deadlines through hard work and determination.

This reflection should be read alongside the e-portfolio, available at [E Portfolio](#)

PART 1 | TEAM PROJECT

Despite personal challenges making it impossible to join the first meeting, I resolved to ensure that I made a contribution to the team. It was particularly helpful that I had worked with one fellow student last year, and felt we had built a good professional relationship. I felt this was a good reflection for the project management team task, as fostering working relationships can lead to successful outcomes. One of my roles in the group was to ensure that the theoretical budget was met ([Model Costingsv3.pdf](#)). This was a challenging task, that led to productive conversations regarding compromise. I was able to draw on my own professional experience of working to and with budgets and imposed constraints. This helped to shape the group response to the task. The group was well balanced and everyone was able to make a contribution. By working together, we were able to stay focused, and produce a piece of work we were all proud of ([Project Report - Synputer.pdf](#)).

This project, was engaging and interesting and it was useful to broaden the subject of software engineering to include hardware engineering. The project was also informative as it considered the development of hardware in the 1980s. In addition to lending a historic perspective, this highlighted the pace of change in the computing landscape, and made me reflect on potential future developments. I have established my own data-driven educational consultancy, and a constant threat to the viability of my own business is to ensure that human, empathetic and wise insights remain ahead of a tide of AI imposters. I can take some reassurance of the emphasis on good human, communication and team work emphasised by Duram (2021).

PART 2 | INDIVIDUAL PROJECT

It was enjoyable to complete a presentation ([Assignment 2 - Slides.pdf](#), [Transcript.pdf](#)), which focused on communication skills and presentation style. I was able to draw on my own experience as an educational leader and consultant, and this compensated for my shallower knowledge around IT hardware and systems. However, I was less motivated to complete this project as I felt it did not develop, or build on the skills traditionally aligned with computing and coding. Perhaps this led to some typos creeping into the presentation. This in itself provided a reminder to stay focused on tasks which may not provide intrinsic motivation. One valuable part of the focus on project management was risk management. I will be adopting the OWASP methodology (Williams, no date) in my own business projects as there is a balance between technical and business risks ([Unit 4 - Risk Models.pdf](#)).

PART 3 | E-PORTFOLIO

The range of formative tasks was varied throughout the module. The nature of project management, overlaps with some social science disciplines, and the focus on the business side of project management did, in my view, overstate jargon, the notion of selling, and promoting, at times, style over substance. In particular, the review of the 'Halo Effect' (Gu et al, 2023) appeared to value visual first impressions of software above functionality ([Factors affecting user experience.pdf](#)). I would normally value functionality over design, but again, this reflection translated to my own working live where I outsource both marketing and web-design.

It was a welcome return to some Python coding near the end of the module that resonated more highly with me. In particular, I found it valuable to apply linters to some of my own code ([Improving Code Quality.pdf](#)). This code is used as a module for some software which my company sells to schools. This meant that the task was relevant to software engineering and the software development lifecycle (SDLC). This highlighted the need to improve the mark-up of my own code and improve adherence to convention. This reminded me of the need to ensure that my own code was readable by others, or indeed myself, to enable modification and updating. My own reflection was that my coding skills have developed as a result of the course and reviewing old code showed my previous lack of understanding of both convention and efficiency. As a result, much of the code I deploy in my own business needs to be refined and become clean code (Anaya, 2021). This is particularly important as growth has meant sharing code with a new business partner.

Overall, the variety in this module and the theme of project management were very valuable in developing my own thinking in a broader sense and considering the application of computing in a real world setting. The parallels and cross-overs with my own professional experience were though provoking. I particularly enjoyed the team element and was able to reflect that complimentary teams allow all to develop while delivering success.

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